**Oluwatosin Ayeni**

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**Wayve, London, UK**

Dear Hiring Team,

I am applying for the Staff ML Performance Engineer (Inference Optimisation) role at Wayve. I bring production-grade experience in GPU inference optimisation — CUDA profiling, KV cache tuning, serving framework benchmarking on H100 hardware — and I am genuinely excited about the inference challenges involved in deploying large end-to-end neural driving models at scale.

Three specific achievements relevant to what Wayve is building:

- **H100 inference benchmarks:** I benchmarked vLLM v0.30 against SGLang v0.5.20 on 2× H100 NVL 94 GB GPUs (NVLink 4.0, CUDA 13.4), Qwen3-8B BF16, TP=2. TTFT, TPOT, and throughput measured across concurrency 1–128. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance. Live: inference-web-theta.vercel.app.
- **CUDA kernel profiling — KV cache bottleneck:** Using NVIDIA Nsight Systems and Compute, I profiled KV cache CUDA operations on RTX 4090 hardware, identified the memory bandwidth bottleneck, and retuned the kernel access pattern to deliver 3.8× throughput improvement. This required reading warp occupancy reports and memory transaction traces directly — not just running benchmarking scripts.
- **Production Kubernetes serving stack:** I built *vllm-k8s-stack* — production LLM serving with vLLM, KV-cache-aware request routing, HPA autoscaling on GPU node pools, and Prometheus/Grafana observability. This is the infrastructure pattern I would bring to Wayve's inference deployment challenges.

Wayve's approach of training one end-to-end neural model rather than modular sensor pipelines is what first caught my attention. The inference throughput and latency requirements for real-time autonomous driving decisions sit exactly at the boundary of model architecture and serving infrastructure that I have been building toward. I am open to relocating to London.

Yours sincerely,

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